SUEZ AI Gateway - Open Source MVP Stack (v2)

Includes Admin UI, Natural Language Policy Builder, and Self-Service Access

Date: March 2026

1. Core Runtime Stack

- Language: Go (Gateway and Control Plane)
- Kubernetes for orchestration
- Helm for packaging and deployment
- NGINX Ingress or Envoy Gateway
- Redis for rate limits and quotas
- PostgreSQL for config and audit (separate DB per environment)

2. Identity and Authentication (MVP supports both modes)

- API Keys stored hashed in PostgreSQL
- OIDC JWT via Keycloak
- Unified IdentityContext for policy and audit
- All requests require authentication

3. Policy Engine

- Open Policy Agent (OPA) as default engine
- Rego-based policies
- OPA sidecar per gateway pod
- Pluggable policy engine interface
- Versioned policy management

4. Admin UI

- React + TypeScript frontend
- Manage tenants, models, tools, and quotas
- Create, review, publish, and rollback policies
- Approve or deny access requests
- All admin actions fully audited

5. Natural Language Policy Builder

- Local open-source LLM via Ollama or vLLM
- Translate plain English policies into Rego
- Generate test cases automatically
- Draft-only mode until human approval
- Diff preview and impact analysis before publish

6. Self-Service Access Portal

- Users authenticate via Keycloak
- Request access to models, tools, API keys, or higher quotas
- Submit justification and environment scope
- Approval workflow with audit trail
- Approved grants enforced via OPA data model

7. Observability

- Prometheus for metrics
- Grafana dashboards
- Loki for logs
- Jaeger with OpenTelemetry for tracing

8. Environment Isolation

- Separate PostgreSQL instances per environment (dev, staging, prod)
- Separate audit storage per environment

- No cross-environment access
- Environment-scoped RBAC enforcement

9. Summary

This MVP delivers centralized AI governance using only open-source technologies. It includes dual authentication, OPA-based policy enforcement, an Admin UI, a natural language policy builder with guardrails, and a self-service access portal with approval workflows.